NUS EES

ENERGY EFFICIENT SOLUTIONS

Making buildings sustainable by saving energy and reducing operational costs



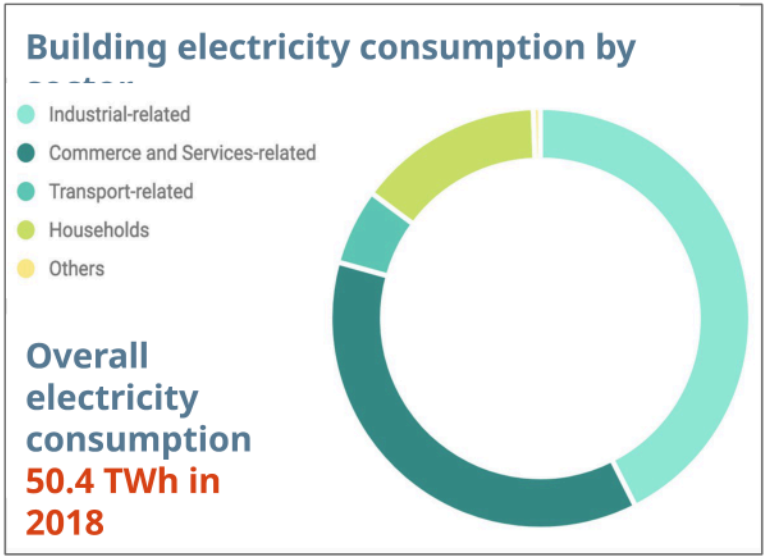
S US, product development, building science



facility engineer, expertise in HVAC and control engineering

Problem

Buildings consume up to 80% of electricity in Singapore*



HVAC (heating, ventilation, and air conditioning) is the largest energy and maintenance expense for building operators

1. HVAC systems run throughout the year in both warm and cool climates
2. HVAC systems have many moving parts that need maintenance and replacement

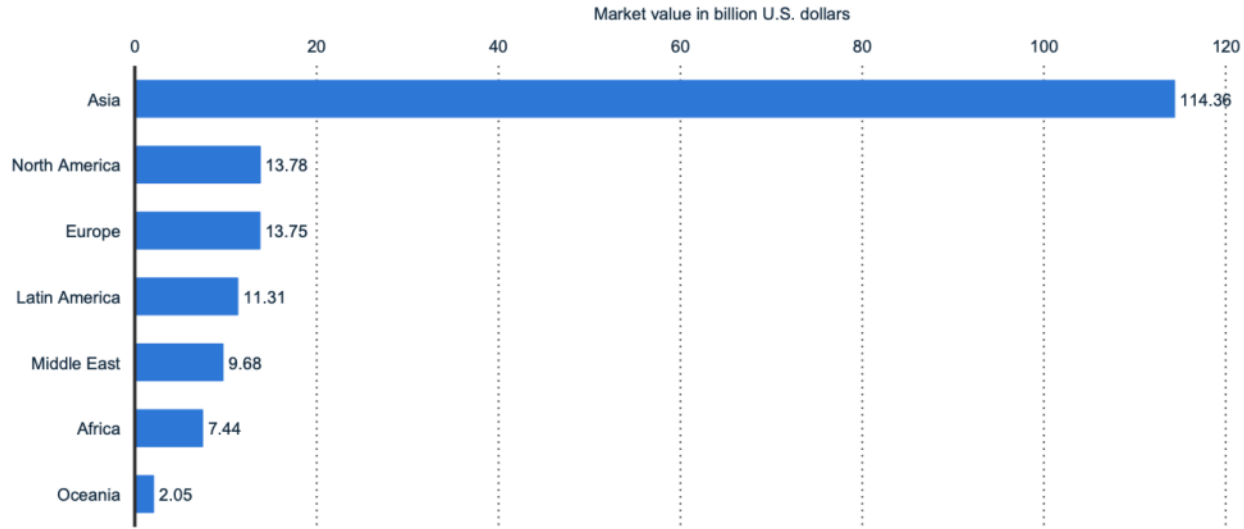


*Source: Energy Market Authority of Singapore

Market: HVAC Industry is >$150 billion*

Market value of heating, ventilation, and air conditioning (HVAC) globally in 2017, by region (in billion U.S. dollars)

Global HVAC market breakdown by region 2017



*Source: Statista

Solution: 3-part Smart Building Product

1. **HVAC Dehumidifier Device**
   - Proprietary desiccant coated bank of tube (DCBT) technology developed by NUS
   - Plug-and-play installation that is easy to maintain
   - Embedded sensors for data trending
   - Overall energy savings of 30%
1. **Digital Management Platform**
   - Premium predictive maintenance software that can be expanded to include other HVAC components
1. **Service-Oriented Business Model**
   - Pay-as-you-Save model for Building Owners and Operators
   - Maintenance and servicing invoiced on top of annualized revenues

Business

NUS
Building Owners
Facility Managers
HVAC Vendors
Manufacturers

Key Activities

Product Engineering
System Simulation
Platform Development

Key Resources

Engineers
Technicians

Value Proposition

End-to-end Solution for Reducing HVAC Energy Consumption and

Relationships

Dedicated Assistance

Channels

Building Owners
Facility Managers

Customer Segments

<u>Commercial</u>
Offices
Hospitals

<u>Industrial</u>
Data Centers
Electronics
Factories
Cold Storage

Summary

Roadmap

IP from NUS ILO
→ Device Design
→ Prototype
Development
→ Real-World Pilot

Competitive Advantage

Proprietary technology
Financial landscape
- Post COVID19 world
- Reducing facility operations cost

Team expertise

Scaling/ Exit Strategy

Expansion of predictive maintenance to other HVAC devices

Acquired by HVAC device company